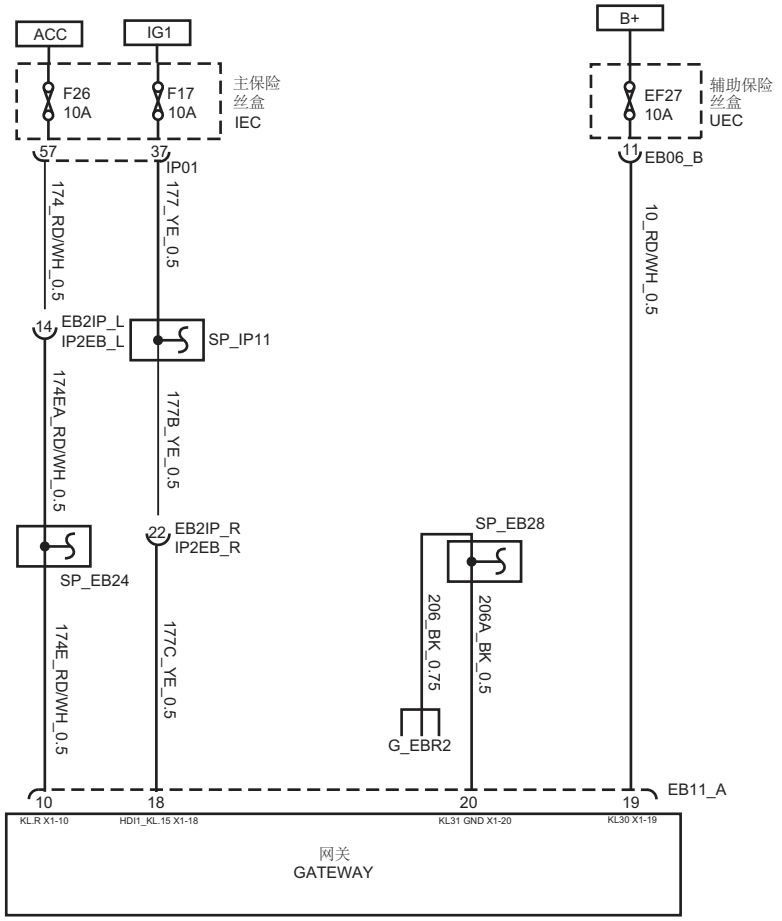
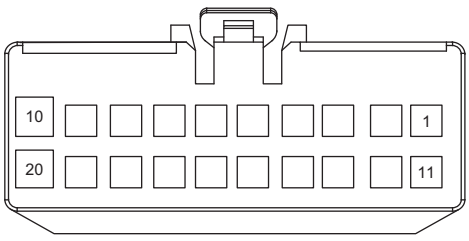
U2001-41、 U2002-42、 B162F-42

Description of fault diagnosis codes
U2001-41：ECU EEPROM Checksum Error
U2002-42：ECU RAM Error
B162F-42：EEPROM(NVM) Error
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • GAW module
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
U2001-41
U2002-42
B162F-42
To set the effect of a fault code condition
1 2
Description of circuit diagram
The GAW module will monitor whether all sensors and actuators are within normal range all the time. Meanwhile,it will check the connector for virtual connection, short circuit, open circuit and the connection rationality with other signals. When fault occurs, the GAW module will save the fault code corresponding to that fault and enable safety mode.

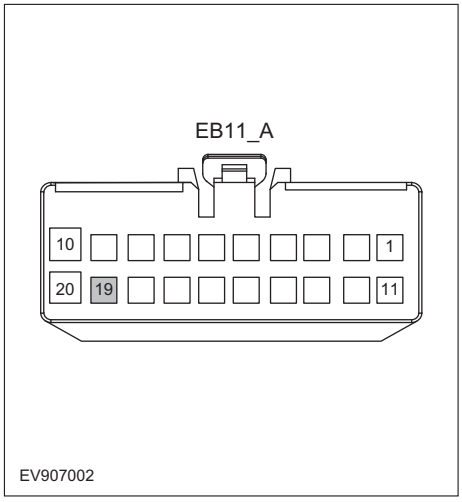
Circuit diagram

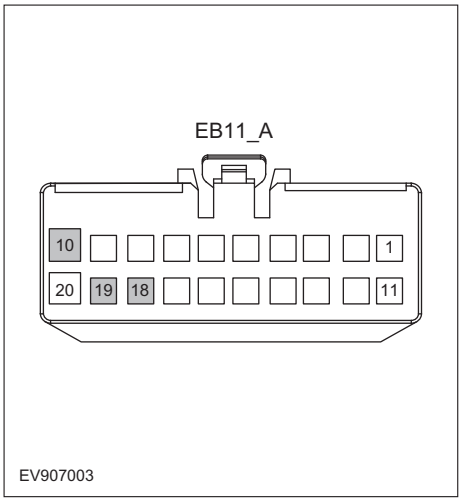
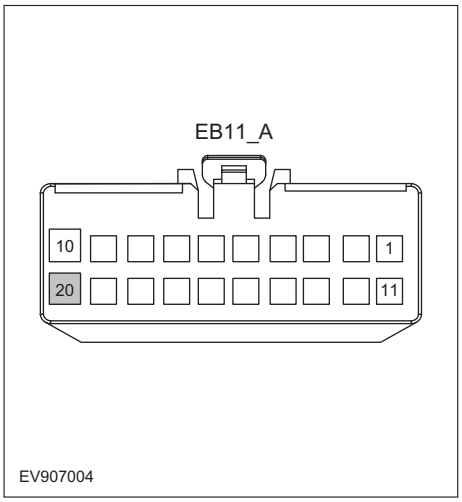


EB11_A



EV907001

Troubleshooting steps	
Step 1	General inspection.
	1 Check fuses EF33/EF34 for damage. 2 Check battery capacity. 3 Check the harness plug of GAW module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the GAW module with diagnostic apparatus. 4 Is there any other fault code except for U2001-41/U2002-42/B162F-42?
	YES → Refer to:DTC Summary list(GAW).
NO ↓	
Step 3	Check the battery of GAW module.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB11_A of network gateway module. 3 Measure the voltage between harness terminal on GAW module and ground. Measuring circuit: voltage between terminal 19 on plug EB11_A and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
	NO → Repair the circuit.
YES ↓	

Step 4	Check the power voltage when the GAW module is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between harness terminal on GAW module and ground. Measuring circuit: voltage between terminal 10 on plug EB11_A and ground. Measuring circuit: voltage between terminal 18 on plug EB11_A and ground. Measuring circuit: voltage between terminal 19 on plug EB11_A and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair the circuit.
YES ↓	
Step 5	Test whether the grounding circuit of GAW module is open.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB11_A of network gateway module. 3 Measure the resistance between harness plug of GAW module and ground. Measuring circuit: resistance between terminal 20 on plug EB11_A and ground. **Standard value: < 1Ω** 4 Test whether the resistance is less than 1Ω?
	NO → Repair the circuit.
YES ↓	

Step 6	Check if GAW module is normal.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB11_A of GAW module. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the GAW module plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Repair GAW module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	